Duyen Thanh Thao Nguyen (myUH ID: 2329157)

Project: Discovering and Predicting Research Topics in Computer Science (DBLP Dataset)

1. Introduction

The rapid growth of computer science research makes it difficult to track new trends and key publications, and the DBLP dataset offers a strong foundation for understanding these developments. This project follows five major stages: preprocessing, exploratory data analysis, clustering, classification and anomaly detection, to uncover research patterns, predict topics or venues, and reveal how computer science evolves over time.

2. Attribute Overview

- **Title:** Concise summary or preview of the article's content.
- **Abstract:** Provides detailed text for feature extraction in clustering and classification.
- **Authors:** Identifies contributors and enables collaboration and productivity analysis.
- **Venue:** Indicates the publication outlet and can serve as a supervised classification label.
- **Year:** Supports temporal analysis of evolving research topics.
- **Citation Count:** Reflects a paper's impact and aids in anomaly or trend detection.
- **References:** Enables citation network and relationship analysis.
- **Unique ID:** Serves as a stable identifier for managing and linking records.

3. Analysis and results

3.1. Preprocessing

1. Load JSON files in 500-row chunks to conserve memory, fill missing title and abstract columns with None and drop records lacking either field.
2. Remove duplicate entries using the (title + abstract) pair as the deduplication key.
3. Standardize text by converting titles and abstracts to lowercase and removing whitespace.
4. Apply a custom cleaning function to the combined title and abstract pairs that strips non-alphabetic characters and collapses extra spaces, producing normalized text for feature extraction.

TF-IDF Feature Extraction

The cleaned text (title + abstract) was vectorized using TF-IDF by removing English stopwords, using only unigrams, limiting to 5000 features, and setting the minimum length of a word to 2 letters. Vectorizer learned the vocabulary from all papers, turned each paper into a large numeric vector based on that vocabulary, and cleaned the vocabularies to make them consistent.

3.2. Exploratory Data Analysis (EDA)

3.2.1. Descriptive Statistics and Citation Statistics

3.2.1.1. Descriptive Statistics

Basic frequency statistics were computed to characterize the dataset and provide an overview of publishing patterns within DBLP:

- Papers per venue: Counted using value_counts() to measure publication volume across venues.
- Papers per year: Sorted yearly counts to show growth trends over time.
- Papers per author: Exploded author lists so each author–paper pair is counted, capturing individual productivity.

3.2.1.2. Citation Statistics

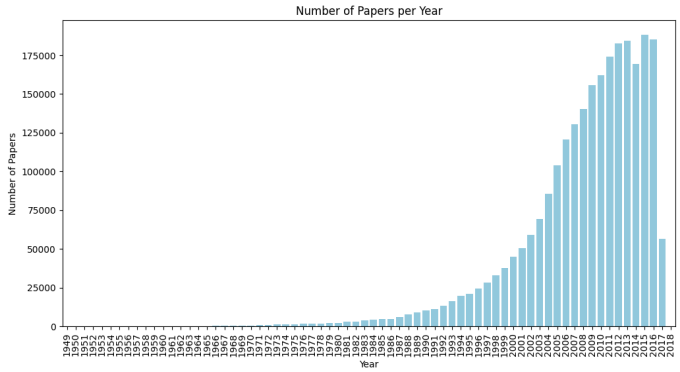
Average citation counts were computed for:

- Average citations per author: measuring researcher impact.
- Average citations per year: capturing annual citation trends.

3.2.2. Distribution Visualizations

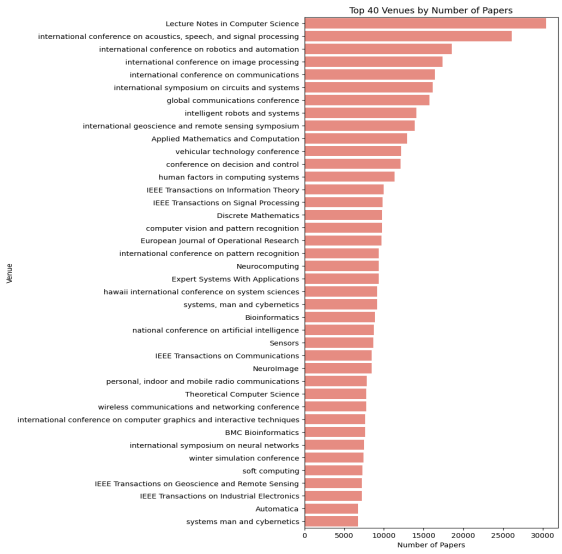
To reveal patterns in publishing activity, citations, and venue characteristics, several visualizations were produced.

Paper Volume Visualization



Number of Papers per Year

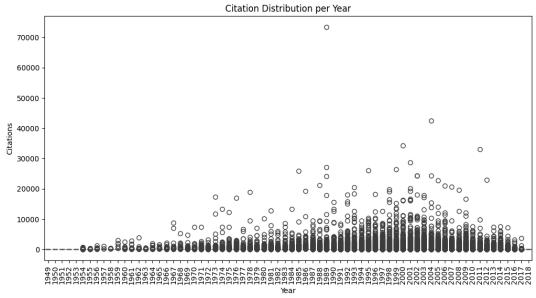
The vertical bar chart shows that publication counts in DBLP grew slowly in the early decades, remained modest until the late 1990s and rose sharply after 2000. The peak appears around 2015–2016, reflecting a period of high research activity. The decline after 2016 is likely due to delayed indexing as newer papers have not yet been fully added to DBLP. Therefore, this drop reflects incomplete records rather than an actual decrease in publications.



Top 40 Venues by Number of Papers

The horizontal bar chart displays the top 40 publication venues in the DBLP dataset, with Lecture Notes in Computer Science (LNCS) appearing far above the rest with almost 30000 papers due to its role in publishing proceedings from many CS conferences. Most of the high-output venues are international conferences in areas such as signal processing, communications, robotics and image processing. Journals and more specialized venues appear mainly in the lower portion of the ranking. Overall, the distribution follows a long-tail pattern in which only a few venues publish extremely large numbers of papers.

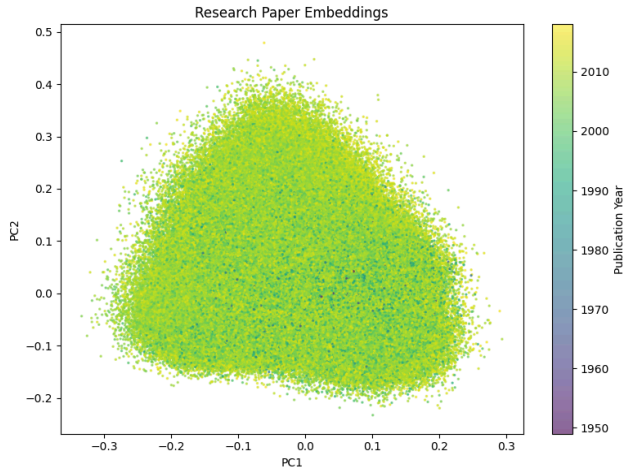
Citation Distribution Boxplots of citation counts by year



The scatter plot illustrates that most papers receive relatively few citations, while a small number become highly influential, including one outlier near 1990 with over 70000 citations. The density of points increases over time, reflecting the growth of scientific publishing. Citation peaks drop sharply after 2014 because newer papers may have had less time to accumulate citations. Overall, the rising spread of citation values indicates that as the research community expanded, the potential impact of highly cited papers increased as well.

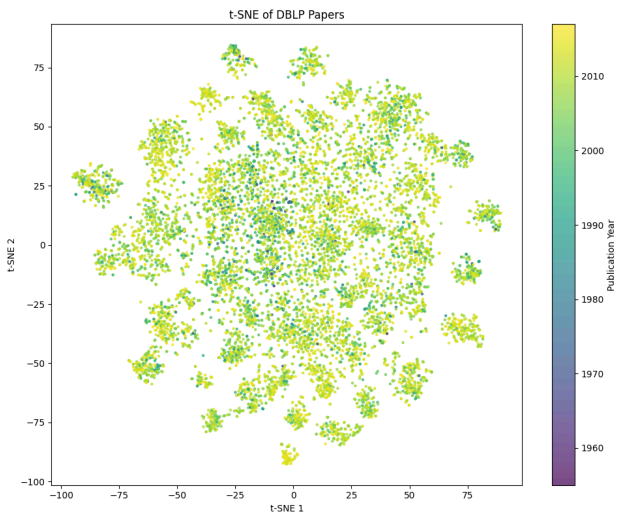
3.2.4. PCA and t-SNE for 2D visualization of paper embeddings

3.2.4.1. PCA



The PCA plot maps papers in two dimensions, colored by publication year, with PC1 and PC2 capturing the strongest text variations. The mixed colors and the single dense cluster indicate that publication year does not drive major differences in the embeddings. Instead, research themes evolve gradually with significantly shared vocabulary, meaning that two principal components are not sufficient to separate them.

3.2.4.2 t-SNE



The points are separated into many separate, small and loose clusters of different sizes, which is expected when visualizing high-dimensional text data. This can suggest that papers are grouped by topical similarity, with newer papers (green, yellow) tending to form distinct and widespread clusters. Also, it mainly shows that the dataset contains diverse papers that do not separate cleanly in two dimensions.

3.2.5. Keyword–Year Analysis and Emerging Research Topics

To study how research themes evolve over time, two keyword-based analyses were performed.

3.2.5.1 Keyword Frequency Matrix

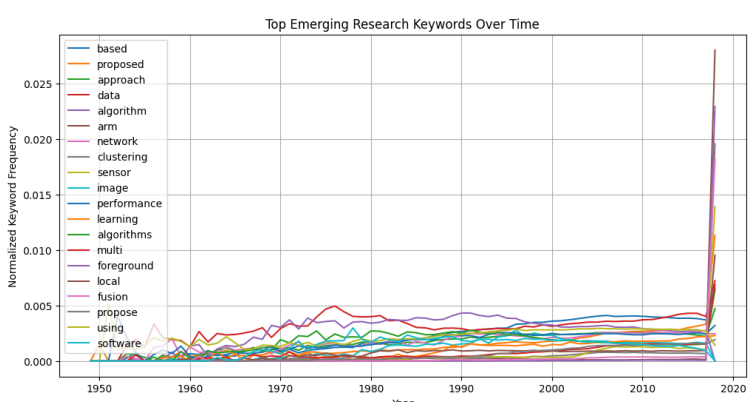
For each year:

The global TF-IDF model was applied to that year's documents.

TF-IDF scores were summed to produce a keyword frequency vector.

3.2.5.2 Trend Detection (Emerging Topics)

To detect emerging research topics: The methodology involved calculating the annual Normalized Term Frequency-Inverse Document Frequency (TF-IDF) for all keywords across the corpus. Subsequently, Simple Linear Regression was applied to these time series to quantify each keyword's emergence rate (slope) and the 20 keywords exhibiting the steepest positive trends were selected and plotted.



The sharp, significant rise of words like ‘network’, ‘learning’, ‘algorithms’ and ‘data’ after 2010 confirms one dominant trend, which

is the explosive growth of AI and Data Science. These keywords are interconnected, showing a massive increase in research focused on building and using complex algorithms and neural networks for machine learning with large datasets.

3.3. Clustering

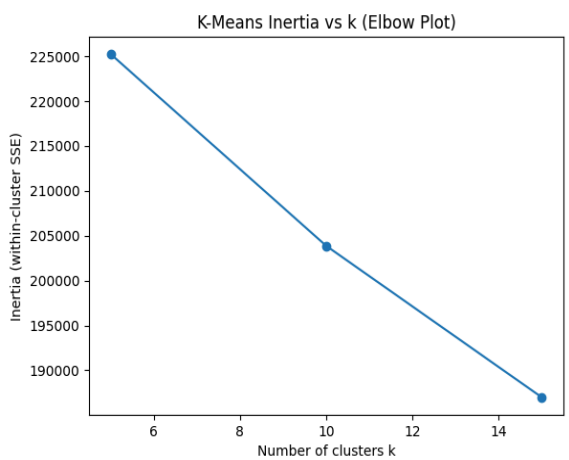
3.3.1. K-Means Clustering on 50D SVD Features with Multiple k Values

K-Means clustering was applied to the 50-dimensional SVD-reduced TF-IDF features using k = 5, 10, and 15. Each model was evaluated using inertia (SSE), runtime, and cluster size distribution.

	k	inertia	fit_time_sec	cluster_sizes
0	5	225219.162931	32.289997	{0: 260523, 1: 228806, 2: 749363, 3: 192199, 4...
1	10	203856.334953	38.335069	{0: 193217, 1: 153618, 2: 923194, 3: 105808, 4...
2	15	186987.137589	57.937495	{0: 40565, 1: 379954, 2: 32177, 3: 108397, 4: ...

Increasing k lowers inertia but with diminishing returns and higher runtime. All settings show cluster imbalance, reflecting dominant research areas, while higher k creates finer but smaller, potentially noisy clusters. Overall, k = 10 offers a good balance of compactness, runtime, and interpretability.

Elbow Plot



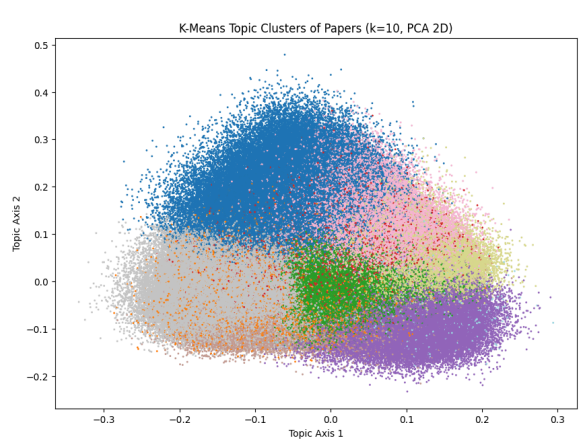
Number of papers per cluster

Fitting final K-Means with k = 10 on all papers…

Cluster ID	Count
0	193217
1	153618
2	923194
3	105808
4	171094
5	73006
6	180338
7	472988
8	241443
9	33247

The inertia curve for k = 5, 10, 15 shows that inertia decreases as k increases, but the rate of improvement drops after k = 10. This flattening indicates an elbow, meaning additional clusters give diminishing returns. Therefore, k = 10 represents the best balance between simplicity and clustering quality.

3.3.2. 2D Visualization of K-Means Topic Clusters (PCA Projection)



The scatter plot shows the 2D PCA projection of the 50-dimensional SVD embeddings, colored by the 10 K-Means clusters. The clusters form broad overlapping but distinguishable regions, indicating that research papers organize into topic groups even after dimensionality reduction. Some clusters appear

dense and compact, suggesting highly coherent topics while others are more spread out, reflecting broader or more mixed research areas. The visual separation between color regions confirms that k = 10 captures major thematic structures within the dataset, even though natural overlap exists due to interdisciplinary papers and shared terminology across fields.

3.3.3. Final 10 Clusters with TopicsTest on a sample of 500 papers:

Cluster ID	Top Keywords	Cluster/ Topic Name
0	network, networks, wireless, routing, sensor, nodes, traffic, protocol, mobile, energy, node, based, performance, communication, packet	Networking & Wireless Communication
1	data, mining, based, analysis, information, model, using, time, approach, clustering, method, paper, database, query, large	Data Mining & Databases
2	model, method, based, using, time, paper, results, analysis, models, approach, used, new, systems, performance, proposed	Machine Learning & Modeling
3	control, robot, controller, robots, systems, model, based, time, feedback, proposed, design, motion, paper, nonlinear, using	Robotics & Control Systems
4	image, images, method, segmentation, based, video, proposed, using, detection, recognition, object, features, color, algorithm, feature	Computer Vision & Image Processing
5	learning, students, based, machine, data, model, training, knowledge, approach, student, algorithm, paper, using, classification, teaching	Education & Learning Analytics
6	power, channel, frequency, performance, interference, signal, proposed, noise, rate, scheme, channels, energy, low, mimo, error	Signal Processing & Communications
7	software, information, web, systems, design, user, based, service, development, research, services, users, paper, knowledge, management	Software Engineering & Web Systems
8	algorithm, problem, graph, algorithms, graphs, problems, optimization, time, search, number, optimal, solution, set, based, scheduling	Algorithms, Graph Theory & Optimization
9	fuzzy, based, logic, proposed, method, model, control, sets, decision, controller, algorithm, rules, membership, systems, approach	Fuzzy Systems & Decision Making

3.3.4. Cluster Quality Evaluation Sample Selection

A random sample of 10,000 points was used to efficiently evaluate K-Means performance by calculating Silhouette Score and Davies-Bouldin Index.

- Silhouette Score: 0.05052727237380478 (measuring point-cluster fit)
- Davies–Bouldin Index: 2.876608654998691 (cluster compactness and separation)

These metrics confirm that the k=10 clustering, while mathematically grouping the data, does not produce high-quality, discrete topics. The clusters are likely highly mixed, reflecting the complex, interrelated nature of the research keywords in the DBLP dataset, which makes clean separation difficult.

3.3.5. Agglomerative Clustering Visualization on PCA-Reduced Subset

	linkage	n_clusters	silhouette	davies_bouldin
0	ward	10	0.089453	2.471949
1	average	10	0.295098	0.942445
2	complete	10	0.109397	2.342668

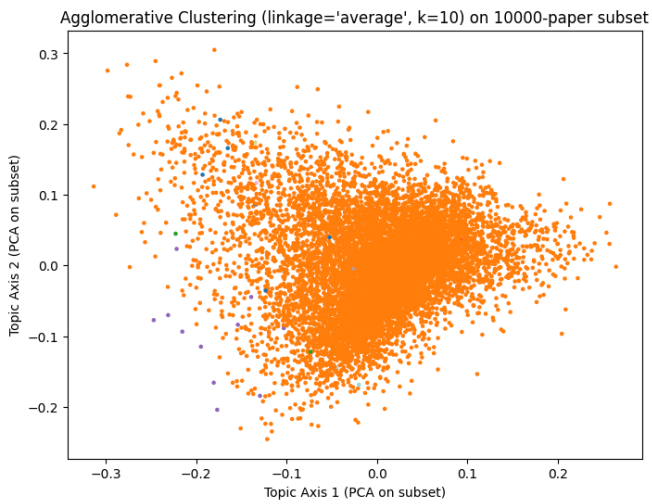
The code runs Agglomerative Clustering (a hierarchical clustering method) using three linkage types:

- **Ward:** minimizes variance within clusters.
- **Average:** uses average distance between clusters.

- **Complete:** uses maximum distance between clusters.

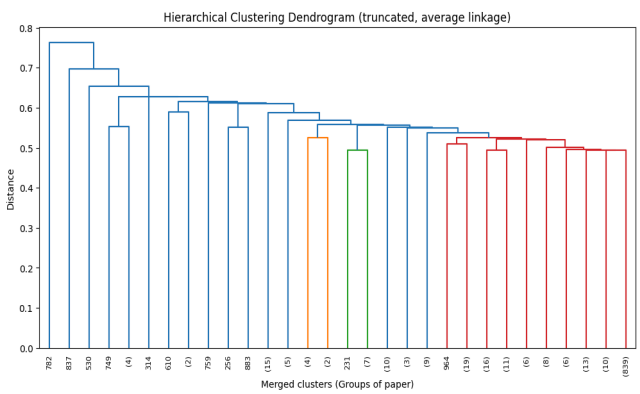
Average linkage:

- Silhouette: 0.295 => noticeably higher than the other methods, indicating better separation of clusters.
- Davies–Bouldin: 0.94 => much lower than the others, showing clusters are more compact and distinct.
- **Conclusion:** This method gives the clearest and most meaningful clustering for the dataset.



The plot shows that Agglomerative Clustering with average linkage groups almost all papers into a single dominant cluster, with only a few scattered points assigned to other groups. This indicates that the algorithm does not find strong or well-separated topic structures in the PCA - reduced space. Overall, although average linkage achieves the best evaluation score, the clusters still heavily overlap, suggesting weak differentiation between papers under this method.

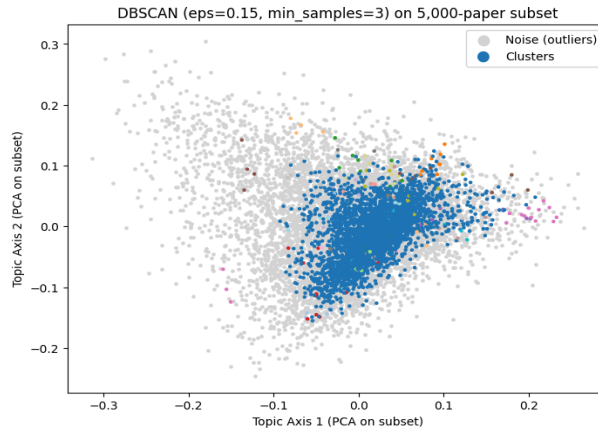
3.3.6. Hierarchical Clustering Dendrogram



The truncated dendrogram illustrates how papers are gradually merged into clusters using average linkage. Most merges occur at relatively similar distances, indicating weak separation and no clearly distinct hierarchical structure among the papers. The evenly spaced branch heights suggest that the dataset lacks strong, well-defined cluster boundaries under this method.

3.3.7. DBSCAN

	eps	min_samples	n_clusters	n_noise
0	0.05	3	3	9975
1	0.05	5	2	9981
2	0.10	3	25	9282
3	0.10	5	4	9415
4	0.15	3	51	6689
5	0.15	5	9	7014
6	0.20	3	19	3084
7	0.20	5	8	3296

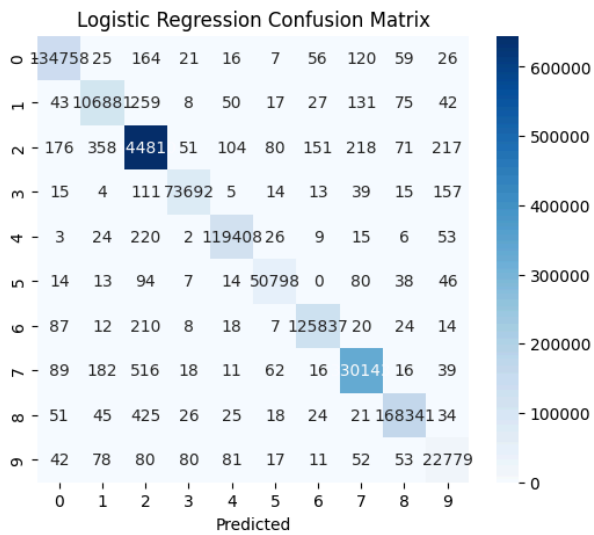


The table shows that very small eps values (0.05–0.10) produce a relatively small number of clusters while labeling the significant majority of papers as noise, indicating that the algorithm is too strict at those settings. However, increasing eps to 0.20 produces the opposite effect, merging many points together and reducing meaningful structure. Thus, eps = 0.15 provides the best balance, reducing noise substantially while identifying a more reasonable number of clusters. This is reflected in the scatter plot, where DBSCAN with eps = 0.15 and min_samples = 3 reveals one main cluster (blue) along with several smaller ones against a background of outliers (gray), suggesting an acceptable level of granularity and noise classification for this text dataset.

3.4. Classification

	Accuracy	Macro-F1	Micro-F1	Precision	Recall
Logistic Regression	0.9965681151	0.9942421190	0.9965681151	0.9942019345	0.9942845035
SVM	0.9507773182	0.9547253878	0.9507773182	0.9581522987	0.9516115577
Random Forest	0.9470393054	0.9481999927	0.9470393054	0.9581863063	0.9387769405
LightGBM	0.9686437522	0.9675682645	0.9686437522	0.9690143791	0.9661379487

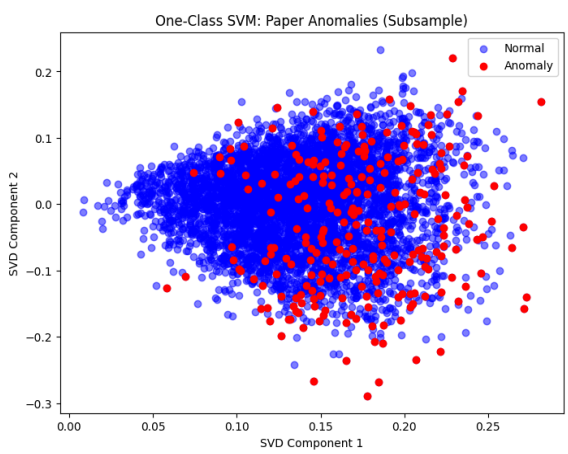
The Logistic Regression model is the clear best performer: its Accuracy (0.9966), Micro-F1 (0.9966), and Macro-F1 (0.9942) are all exceptionally high, showing it is correct 99.66% of the time and performs consistently well across all venues. Its Recall (0.9943) is slightly higher than its Precision (0.9942), meaning it is marginally better at avoiding False Negatives (missing a paper's true venue) than False Positives (assigning the wrong venue). In contrast, the Random Forest performs the worst, with Accuracy 0.9470 and the lowest Recall (0.9388), indicating its non-linear approach leads to the most missed or unclassified papers.



The confusion matrix shows that the Logistic Regression model predicts venues with extremely high accuracy and very few mistakes. This means the SVD features and supervised classifier successfully capture the underlying structure of research topics. With such reliable predictions, the project can confidently assign topics/venues to new papers, compare topic distributions across fields, and identify patterns in scientific publishing. In other words, the strong performance seen in the matrix confirms that your supervised approach works well and directly supports the project's goal of uncovering research themes and accurately labeling new papers.

3.5. Anomaly Detection

The code uses a One-Class SVM (OCSVM) with an RBF kernel and nu = 0.05 to detect anomalous research papers within a 5,000 document subsample by learning a boundary around the "normal" SVD features. After flagging about 5% as outliers, the results are validated by examining each outlier's top TF-IDF keywords, title, and venue, linking the statistical anomaly to its unusual research content and confirming the deviation on a 2D scatter plot.



Output: Detected 255 anomalous papers in the subsample.

This plot shows how the One-Class SVM separated normal papers (blue) from anomalous ones (red) using two SVD features. Most points cluster tightly in the center, meaning their content looks typical for the dataset. The red points spread more loosely around the edges, showing that these papers have unusual feature patterns compared to the main group. In simple terms: the blue dots represent papers whose writing/content style fits the norm, while the red dots are those whose language patterns stand out as different enough to be flagged as anomalies.

4. Conclusion

Our project successfully met its goal of uncovering underlying research topics and predicting the topic or venue of new papers through a combination of unsupervised and supervised learning techniques. Using TF-IDF and SVD to embed millions of titles and abstracts, we revealed clear patterns in scientific

publishing, including major topic clusters, shared vocabulary across venues, and long-term shifts in research focus. Unsupervised methods such as K-Means, hierarchical clustering, and DBSCAN helped map the thematic structure of the DBLP dataset, while trend analysis identified emerging areas like machine learning and data-driven research.

Supervised classification, especially Logistic Regression, achieved extremely high accuracy in predicting venues, demonstrating that the learned text features capture the distinctive language of each field. Anomaly detection further highlighted papers with unusual or niche content, offering additional insight into the diversity of research topics. Overall, these methods show that large-scale textual analysis can effectively organize scientific literature, reveal meaningful topic relationships, and reliably classify new research, fulfilling the project's central objective.

5. Challenges Encountered

- **High Dimensionality of Text Data:** Research papers contain extensive vocabulary and technical terminology, resulting in extremely high-dimensional TF-IDF vectors. Reducing this dimensionality while preserving semantic meaning was a key difficulty.
- **Choosing the Optimal Number of Topics/Clusters:** Methods like K-means and LDA require selecting parameters. Finding values that produced coherent, interpretable results without overfitting was challenging.
- **Noise and Inconsistency in Real-World Data:** The dataset contained variations in writing style, terminology, abbreviations, and fields, making it difficult for models to distinguish meaningful patterns from noise.
- **Computational Cost:** Processing millions of papers, performing SVD, and running clustering models required significant computational resources and optimization.
- **Balancing Unsupervised and Supervised Approaches:** Integrating the results from clustering (unsupervised) with classification (supervised) needed careful evaluation to ensure both approaches complemented each other and improved prediction accuracy.
- **Interpreting Model Outputs:** Even once models produced topics or predictions, translating them into human-readable insights, such as labeling clusters or identifying characteristic keywords, posed interpretability challenges.